

FEETECH STS3215

Serial Bus Smart Servo Complete Reference Manual (English)

7.4V / 19 kg.cm · TTL Serial Bus · 12-bit Magnetic Encoder · 360°

Manufacturer	Shenzhen FEETECH RC Model Co., Ltd.
Model	STS3215 / ST-3215-C001 (7.4V) · ST-3215-C018 (12V)
Series	STS — TTL Serial Bus, SMS1.0 Protocol
Document compiled	February 2026
Sources	Official FEETECH datasheets, communication protocol manual, memory table (magnetic encoding version)

1. Product Overview

The STS3215 is a high-performance TTL serial bus smart servo from FEETECH. It uses a 12-bit magnetic coding sensor providing 360° absolute position control with 4096 steps of resolution. Unlike potentiometer-based servos, the magnetic encoder has no mechanical wear and offers virtually unlimited rotation. Multiple operating modes are supported: position servo, closed-loop speed (motor), open-loop speed, and stepper. Up to 253 servos can be daisy-chained on a single bus.

2. Electrical & Mechanical Specifications

Parameter	@ 6V	@ 7.4V	Notes
Typical operating voltage	6 V	7.4 V	Rated range: 4 V – 7.4 V
No-load speed	0.238 s/60° (42 RPM)	0.192 s/60° (52 RPM)	
No-load current	130 mA	150 mA	
Stall torque	16.5 kg.cm	19.5 kg.cm	229 / 271 oz.in
Stall current	2 A	2.5 A	
Idle current (stopped)	21 mA	21 mA	
Rated torque	4 kg.cm	5 kg.cm	
Rated current	500 mA	650 mA	
Motor resistance	2.5 Ω	—	
Kt constant	7.8 kg.cm/A	—	

Parameter	Value
Dimensions	45.2 × 24.7 × 35 mm
Weight	55 ± 1 g
Case material	PA + GF (glass-filled nylon)
Gear material	Copper (metal gearbox)
Gear ratio	1 : 345
Bearing type	Ball bearings
Output shaft	25T spline / OD 5.9 mm
Backlash	≤ 0.5°
Angle sensor	12-bit magnetic, 360°, unlimited life
Connector	5264-3P, PP material, 15 cm cable
Pin 1 / 2 / 3	GND / Vcc / Signal (TTL)
Operating temperature	-20 °C to +60 °C
Storage temperature	-30 °C to +80 °C
Waterproofing	None
Reliability (life test)	> 100 000 cycles at 1/5 stall torque

3. Control Characteristics

Parameter	Value	Notes
Communication signal	Digital packet	TTL half-duplex, single wire
Protocol type	Half-duplex async serial	8-bit, 1 stop, no parity
ID range	0 – 253	Broadcast ID = 254 (0xFE)
Default baud rate	1 000 000 bps	Configurable: 38400 – 1 Mbps
Control algorithm	PID	P/I/D gains are user-configurable
Neutral position	180° = 2048 counts	Centre of 0–4095 range
Angular range	0 – 360° (0 – 4095 counts)	Full-turn absolute in servo mode
Resolution	0.088° per count	360° / 4096
Rotation direction	Clockwise: 0 → 4096	Configurable
Feedback	Position, speed, load, voltage, current, temperature	
Max position update rate	1 ms	
Signal high voltage	2 V – 5 V	
Signal low voltage	0.0 V – 0.45 V	
Multi-turn mode	±7 turns absolute (lost on power-off)	Resolution can be doubled

3.1 Operating Modes

Mode	Register 33 value	Description
Position servo (default)	0	Absolute position control, 0–360°. Default mode.
Closed-loop speed (motor)	1	Motor mode. Speed is maintained regardless of load increase.
Open-loop speed (motor)	2	Motor mode. Speed decreases as load increases.
Stepper	3	Move relative to current position by a defined step.

3.2 Electronic Protections

Protection	Default threshold	Behaviour
Overload	> 80% of stall torque for 2 s	Reduces torque output. Cleared on next position command. Customizable.
Overcurrent	> 2 A for 2 s	Disables output. Cleared on next position command.
Overvoltage	> 8 V or < 4 V	Enters protection. Automatically released when voltage returns to normal range.
Overtemperature	> 70 °C	Disables torque output.

4. Communication Protocol (SMS/STS Series)

The STS3215 uses the SMS 1.0 half-duplex TTL serial protocol. **IMPORTANT:** In the SMS/STS series, multi-byte values are transmitted **low byte first, high byte second** (little-endian). This is the opposite of the SCS series (big-endian).

4.1 Instruction Packet Format

Byte	Field	Value / Description
1	Header	0xFF
2	Header	0xFF
3	ID	Servo ID (0x00–0xFD). Broadcast = 0xFE
4	Length	Number of parameters N + 2
5	Instruction	See instruction codes table below
6..N+5	Parameters	Command-specific data
N+6	Checksum	~(ID + Length + Instruction + Param1 + ... + ParamN) & 0xFF

4.2 Instruction Codes

Instruction	Code	Param length	Description
PING	0x01	0	Query servo operating status
READ DATA	0x02	2	Read from memory table (address, length)
WRITE DATA	0x03	≥ 1	Write to memory table
REG WRITE	0x04	≥ 2	Buffered write — executes on ACTION command
ACTION	0x05	0	Execute all pending REG WRITE commands
SYNC WRITE	0x83	≥ 2	Write to multiple servos simultaneously (broadcast)
RESET	0x06	0	Reset memory table to factory defaults

4.3 Reply Packet Format

The servo replies with: **0xFF 0xFF ID Length ERROR [Data...] Checksum**. ERROR = 0 means no error. For READ DATA replies, the data bytes follow ERROR.

4.4 Communication Examples

Action	Packet (hex)
PING servo ID 1	FF FF 01 02 01 FB
Read present position (2 bytes at addr 0x38), ID 1	FF FF 01 04 02 38 02 BE
Move ID 1 to pos 2048 at speed 1000 (6 bytes at addr 0x2A: pos L/H, time L/H, speed L/H)	FF FF 01 09 03 2A 00 08 00 00 E8 03 D5
Enable torque, ID 1 (write 1 at addr 0x28)	FF FF 01 04 03 28 01 CE
Disable torque, ID 1 (write 0 at addr 0x28)	FF FF 01 04 03 28 00 CF
Reset to factory, ID 1	FF FF 01 02 06 F6

5. Complete Memory Table (SMS/STS — Magnetic Encoding Version)

Multi-byte values: **low byte first (little-endian)**. Access: R = Read-only, R/W = Read-Write. EEPROM registers require Torque OFF + LOCK=0 before writing, then LOCK=1 after. SRAM registers can be written at any time (torque state independent unless noted).

5.1 EEPROM (non-volatile — read-only area)

Addr (dec)	Addr (hex)	Name	Bytes	Access	Default	Description
3	0x03	MODEL_NUMBER_L	1	R	—	Model number low byte
4	0x04	MODEL_NUMBER_H	1	R	—	Model number high byte

5.2 EEPROM (non-volatile — read/write area)

Note: To write these registers: disable torque (addr 40 = 0), set LOCK=0 (addr 55 = 0), write value, then set LOCK=1 (addr 55 = 1).

Addr	Hex	Name	Bytes	Range	Default	Description
5	0x05	ID	1	0–253	1	Servo ID on the bus
6	0x06	BAUD_RATE	1	0–7	0 (=1Mbps)	0=1M, 1=500k, 2=250k, 3=128k, 4=115.2k, 5=76.8k, 6=57.6k
7	0x07	RETURN_DELAY	1	0–254	0	Reply delay in units of 2 μ s
8	0x08	STATUS_RETURN_LEVEL	1	0–2	1	0=No reply, 1=Reply to read only, 2=Reply to all
9	0x09	MIN_ANGLE_LIMIT_L	1	—	0	Minimum angle limit — low byte
10	0x0A	MIN_ANGLE_LIMIT_H	1	—	0	Minimum angle limit — high byte (0 = no limit)
11	0x0B	MAX_ANGLE_LIMIT_L	1	—	0	Maximum angle limit — low byte
12	0x0C	MAX_ANGLE_LIMIT_H	1	—	0	Maximum angle limit — high byte (0 = no limit)
13	0x0D	MAX_TEMPERATURE	1	0–100	70	Temperature protection threshold ($^{\circ}$ C)
14	0x0E	MAX_VOLTAGE	1	0–254	80 (=8.0V)	Max voltage threshold $\times$ 0.1 V
15	0x0F	MIN_VOLTAGE	1	0–254	40 (=4.0V)	Min voltage threshold $\times$ 0.1 V
16	0x10	MAX_TORQUE_L	1	—	—	Maximum torque — low byte
17	0x11	MAX_TORQUE_H	1	0–1000	1000	Maximum torque — high byte (1000 = 100%)
18	0x12	PHASE	1	—	—	Phase setting (motor direction)
19	0x13	UNLOADING_CONDITION	1	bitmask	—	Protection enable bits: bit5=overload, bit2=temp, bit0=voltage
20	0x14	LED_ALARM_CONDITION	1	bitmask	—	LED alarm condition bits (same mapping as addr 19)
21	0x15	P_COEFFICIENT	1	0–254	32	PID proportional gain
22	0x16	D_COEFFICIENT	1	0–254	32	PID derivative gain
23	0x17	I_COEFFICIENT	1	0–254	0	PID integral gain
24	0x18	MIN_STARTUP_FORCE_L	1	—	—	Minimum startup torque — low byte
25	0x19	MIN_STARTUP_FORCE_H	1	0–1000	—	Minimum startup torque — high byte
26	0x1A	CW_DEAD_BAND	1	0–32	0	Clockwise dead band (position error tolerance)
27	0x1B	CCW_DEAD_BAND	1	0–32	0	Counter-clockwise dead band
28	0x1C	PROTECTION_CURRENT_L	1	—	—	Overcurrent protection threshold — low byte
29	0x1D	PROTECTION_CURRENT_H	1	0–511	—	Overcurrent protection threshold — high byte (unit: $\sim$ 6.5 mA)
30	0x1E	ANGULAR_RESOLUTION	1	1–100	1	Multi-turn resolution multiplier
31	0x1F	OFFSET_L	1	—	0	Homing offset — low byte
32	0x20	OFFSET_H	1	—	0	Homing offset — high byte (signed, $\pm$ 2047)
33	0x21	OPERATING_MODE	1	0–3	0	0=position, 1=speed closed-loop, 2=speed open-loop, 3=speed torque
34	0x22	PROTECTION_TIME_L	1	—	—	Overload protection time — low byte (unit: 40 ms)

35	0x23	PROTECTION_TIME_H	1	—	250	Overload protection time — high byte
36	0x24	OVERLOAD_TORQUE	1	0–100	80	Overload torque threshold as % of max torque
37	0x25	SPEED_CLOSED_LOOP_P	1	0–254	10	Speed closed-loop P coefficient
38	0x26	OVER_CURRENT_TIME	1	0–254	—	Overcurrent protection duration (unit: 40 ms)
39	0x27	VELOCITY_CLOSED_LOOP_I	1	0–254	—	Speed closed-loop I coefficient

5.3 SRAM (volatile — read/write area)

SRAM registers are lost on power-off. They can be written freely (no LOCK/torque precautions needed unless noted).

Addr	Hex	Name	Bytes	Range	Default	Description
40	0x28	TORQUE_ENABLE	1	0–1	0	0 = torque off (servo free), 1 = torque on. Write 128 to set.
41	0x29	ACCELERATION	1	0–254	0	Acceleration ramp (0 = max acceleration). Unit: 100 steps/s ² .
42	0x2A	GOAL_POSITION_L	1	—	—	Target position — low byte
43	0x2B	GOAL_POSITION_H	1	0–4095	—	Target position — high byte
44	0x2C	GOAL_TIME_L	1	—	0	Time to reach target — low byte (0 = use goal speed)
45	0x2D	GOAL_TIME_H	1	0–65535	0	Time to reach target — high byte (unit: ms)
46	0x2E	GOAL_SPEED_L	1	—	0	Target speed — low byte
47	0x2F	GOAL_SPEED_H	1	0–32767	0	Target speed — high byte (unit: steps/s). 0 = max speed.
48	0x30	TORQUE_LIMIT_L	1	—	—	Torque limit — low byte (runtime override)
49	0x31	TORQUE_LIMIT_H	1	0–1000	1000	Torque limit — high byte (1000 = 100%)
50	0x32	LOCK	—	—	—	(alias — see addr 55)
55	0x37	LOCK	1	0–1	1	EEPROM lock: 0 = unlocked (allows EEPROM writes), 1 = locked.

5.4 SRAM (volatile — read-only area)

Addr	Hex	Name	Bytes	Format	Description
56	0x38	PRESENT_POSITION_L	1	—	Current position — low byte
57	0x39	PRESENT_POSITION_H	1	15-bit signed	Current position — high byte. Bit 15 = sign (sign-magnitude encoding). Bit 16 = direction.
58	0x3A	PRESENT_SPEED_L	1	—	Current speed — low byte
59	0x3B	PRESENT_SPEED_H	1	15-bit signed	Current speed — high byte. Bit 15 = direction.
60	0x3C	PRESENT_LOAD_L	1	—	Current load — low byte
61	0x3D	PRESENT_LOAD_H	1	10-bit signed	Current load — high byte. Range 0–1000 (1000 = 100% max load). Bit 16 = direction.
62	0x3E	PRESENT_VOLTAGE	1	uint8	Current input voltage × 0.1 V (e.g. 74 = 7.4 V)
63	0x3F	PRESENT_TEMPERATURE	1	uint8	Current temperature in °C
64	0x40	REGISTERED_INSTRUCTION	1	0–1	1 = a REG WRITE instruction is pending
65	0x41	ERROR_STATUS	1	bitmask	Current error/protection status flags
66	0x42	MOVING	1	0–1	1 = servo is currently moving
69	0x45	PRESENT_CURRENT_L	1	—	Current draw — low byte
70	0x46	PRESENT_CURRENT_H	1	15-bit signed	Current draw — high byte. Unit ≈ 6.5 mA. Bit 15 = direction.

6. Quick Reference — Common Operations

6.1 Key Register Summary

Address	Name	Type	Most Important Use
5	ID	EEPROM R/W	Change servo ID on the bus
6	BAUD_RATE	EEPROM R/W	Change communication speed
9–12	MIN/MAX_ANGLE_LIMIT	EEPROM R/W	Constrain movement range. Set to 0 to disable limits.
21	P_COEFFICIENT	EEPROM R/W	PID stiffness — increase if position lags
22	D_COEFFICIENT	EEPROM R/W	PID damping — increase if overshoot
23	I_COEFFICIENT	EEPROM R/W	PID static error correction
31–32	OFFSET (homing)	EEPROM R/W	Zero-point calibration offset (signed ± 2047)
33	OPERATING_MODE	EEPROM R/W	0=servo, 1=speed CL, 2=speed OL, 3=stepper
40	TORQUE_ENABLE	SRAM R/W	1 = motor on, 0 = free wheel. Write 128 = set centre.
41	ACCELERATION	SRAM R/W	Motion smoothing (0 = instant, higher = smoother ramp)
42–43	GOAL_POSITION	SRAM R/W	Target position 0–4095 (low byte first)
44–45	GOAL_TIME	SRAM R/W	Time in ms to reach target (0 = use GOAL_SPEED)
46–47	GOAL_SPEED	SRAM R/W	Speed in steps/s (0 = max speed)
48–49	TORQUE_LIMIT	SRAM R/W	Runtime torque cap 0–1000 (1000=100%)
55	LOCK	SRAM R/W	0 = unlock EEPROM for writing, 1 = locked
56–57	PRESENT_POSITION	SRAM R	Current position readout
62	PRESENT_VOLTAGE	SRAM R	Input voltage $\times 0.1$ V
63	PRESENT_TEMPERATURE	SRAM R	Temperature in $^{\circ}\text{C}$
66	MOVING	SRAM R	1 = servo in motion
69–70	PRESENT_CURRENT	SRAM R	Current draw $\approx$ value $\times 6.5$ mA

6.2 EEPROM Write Procedure

Step 1	Disable torque	Write 0 to address 40
Step 2	Unlock EEPROM	Write 0 to address 55
Step 3	Write target register	e.g. write 3 to address 33 to set stepper mode
Step 4	Lock EEPROM	Write 1 to address 55
Step 5	Re-enable torque (optional)	Write 1 to address 40

6.3 Baud Rate Codes

Register value (addr 6)	Baud rate
0 (default)	1 000 000 bps
1	500 000 bps
2	250 000 bps
3	128 000 bps
4	115 200 bps
5	76 800 bps
6	57 600 bps

7	38 400 bps
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6.4 Typical Python (scservo_sdk) Snippets

Connect and ping:

```
portH = PortHandler('/dev/ttyACM0')
pktH = PacketHandler(0) # 0 = SMS/STS protocol
portH.openPort()
portH.setBaudRate(1000000)
model, res, err = pktH.ping(portH, 1)
```

Enable torque:

```
pktH.write1ByteTxRx(portH, 1, 40, 1)
```

Move to position 2048 at speed 2000:

```
pktH.write2ByteTxRx(portH, 1, 46, 2000) # set speed first
pktH.write2ByteTxRx(portH, 1, 42, 2048) # then target position
```

Read present position:

```
pos, res, err = pktH.read2ByteTxRx(portH, 1, 56)
```

Fix corrupted limits (Ierobot issue):

```
pktH.write1ByteTxRx(portH, 1, 40, 0) # torque off
pktH.write1ByteTxRx(portH, 1, 55, 0) # unlock EEPROM
pktH.write2ByteTxRx(portH, 1, 9, 0) # min limit = 0 (no limit)
pktH.write2ByteTxRx(portH, 1, 11, 0) # max limit = 0 (no limit)
pktH.write2ByteTxRx(portH, 1, 31, 0) # homing offset = 0
pktH.write1ByteTxRx(portH, 1, 55, 1) # lock EEPROM
pktH.write2ByteTxRx(portH, 1, 46, 2000) # restore speed
```

7. Troubleshooting Guide

Symptom	Likely cause	Solution
Servo does not move despite command	Torque not enabled Goal speed = 0 Angle limits too narrow	Write 1 to addr 40 Write non-zero to addr 46 Write 0 to addrs 9–12 to clear limits
Servo does not respond to bus	Wrong baud rate Wrong ID	Try 1 000 000 bps (default) Broadcast PING to discover ID
Position reads as -100 or abnormal	Calibration values from LeRobot normalised the reading	Use raw register reads (addr 56). Ignore LeRobot calibration layer for diagnosis
Servo stuck at one position	MIN_ANGLE_LIMIT = MAX_ANGLE_LIMIT (1-Reset addrs 9–12 to 0 following EEPROM write procedure)	
Servo overheats / shuts down	Overtemperature protection triggered (> 70°C)	Allow to cool. Check mechanical load and torque limit (addr 48–49).
Servo moves then stops unexpectedly	Overload protection triggered	Reduce load or increase PROTECTION_TIME (addr 34–35). Check OVERCURRENT_LIMIT (addr 48–49).
Cannot write to EEPROM	LOCK register = 1 (or torque enabled)	Disable torque (addr 40 = 0) then write 0 to addr 55 before any EEPROM write
LeRobot corrupted servo registers	LeRobot calibration writes homing offset & angle limits to EEPROM	Use <code>le_robot_erase_eeprom</code> to reset addrs 9–12 (limits) and 31–32 (offset) to 0

8. Additional Resources

Official FEETECH documentation portal	http://doc.feetech.cn
Protocol & memory table (magnetic version) download	http://www.feetechrc.com/letter-of-agreement.html
Datasheet (Seeed Studio)	https://files.seeedstudio.com/products/Feetech/108090023_STS3215-C001_Datasheet.pdf
FD debugging software	https://www.feetechrc.com/Software.html
LeRobot Feetech driver (registers reference)	https://github.com/huggingface/lerobot/blob/main/lerobot/motors/feetech.py
Arduino STS_servos library	https://github.com/matthieuvigne/STS_servos